Taipei Veterans General Hospital

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## Sample Information

**Patient Name:** 邱寶華
**Gender:** Female
**ID No.:** P221196364
**History No.:** 49165073
**Age:** 57

**Ordering Doctor:** DOC3084A 陳胤之
**Ordering REQ.:** 0CDZYEW
**Signing in Date:** 2022/12/29

**Path No.:** M111-00036
**MP No.:** F22134
**Assay:** Oncomine Focus Assay
**Sample Type:** FFPE
**Block No.:** S22-11058A2 from En Chu Kong Hospital
**Percentage of tumor cells:** 70%

**Reporting Doctor:** DOC5466K 葉奕成 (Phone: 8#5466)

**Note:**

## Sample Cancer Type: Breast Cancer

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### Report Highlights

1 Relevant Biomarkers
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## Relevant Breast Cancer Variants

| Gene | Finding |
|---|---|
| ERBB2 | None detected |
| PIK3CA | ***PIK3CA p.(V105_R108del) c.314_325delTAGGCAACCGTG*** |

## Relevant Biomarkers

| Tier | Genomic Alteration | Relevant Therapies (In this cancer type) | Relevant Therapies (In other cancer type) | Clinical Trials |
|---|---|---|---|---|
| **IIC** | *PIK3CA p.(V105_R108del) c.314_325delTAGGCAACCGTG*<br>phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha<br>Allele Frequency: 38.64% | None | None | 3 |

**Public data sources included in relevant therapies: FDA**[1], NCCN, **EMA**[2], ESMO
**Tier Reference:** Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

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# Variant Details

## DNA Sequence Variants

| Gene | Amino Acid Change | Coding | Variant ID | Locus | Allele Frequency | Transcript | Variant Effect | Coverage |
|---|---|---|---|---|---|---|---|---|
| PIK3CA | p.(V105_R108del) | c.314_325delTAGGCAACCGTG | . | chr3:178916925 | 38.64% | NM_006218.4 | nonframeshift Deletion | 1972 |
| JAK1 | p.(P733=) | c.2199A>G | . | chr1:65310489 | 48.55% | NM_002227.4 | synonymous | 1998 |
| ALK | p.(D1529E) | c.4587C>G | . | chr2:29416366 | 49.97% | NM_004304.5 | missense | 1999 |
| ALK | p.(G1125=) | c.3375C>A | . | chr2:29445458 | 48.52% | NM_004304.5 | synonymous | 1997 |
| FGFR4 | p.(P136L) | c.407C>T | . | chr5:176517797 | 99.30% | NM_213647.3 | missense | 2000 |
| EGFR | p.(Q787=) | c.2361G>A | . | chr7:55249063 | 50.73% | NM_005228.5 | synonymous | 1999 |
| MET | p.(N375S) | c.1124A>G | . | chr7:116340262 | 32.32% | NM_001127500.3 | missense | 1999 |

# Biomarker Descriptions

**PIK3CA (phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha)**

Background: The PIK3CA gene encodes the phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha of the class I phosphatidylinositol 3-kinase (PI3K) enzyme[1]. PI3K is a heterodimer that contains a p85 regulatory subunit, which couples one of four p110 catalytic subunits to activated tyrosine protein kinases[2,3]. The p110 catalytic subunits include p110α, β, δ, γ and are encoded by genes PIK3CA, PIK3CB, PIK3CD, and PIK3CG, respectively[2]. PI3K catalyzes the conversion of phosphatidylinositol (4,5)-bisphosphate (PI(4,5)P2) into phosphatidylinositol (3,4,5)-trisphosphate (PI(3,4,5)P3) while the phosphatase and tensin homolog (PTEN) catalyzes the reverse reaction[4,5]. The reversible phosphorylation of inositol lipids regulates diverse aspects of cell growth and metabolism[4,5,6,7]. Recurrent somatic alterations in PIK3CA are frequent in cancer and result in the activation of PI3K/AKT/MTOR pathway, which can influence several hallmarks of cancer including cell proliferation, apoptosis, cancer cell metabolism and invasion, and genetic instability[8,9,10].

Alterations and prevalence: Recurrent somatic activating mutations in PIK3CA are common in diverse cancers and are observed in 20-30% of breast, cervical, and uterine cancers and 10-20% of bladder, gastric, head and neck, and colorectal cancers[11,12]. Activating mutations in PIK3CA commonly occur in exons 10 and 21 (previously referred to as exons 9 and 20 due to exon 1 being untranslated)[13,14]. These mutations typically cluster in the exon 10 helical (codons E542/E545) and exon 21 kinase (codon H1047) domains, each having distinct mechanisms of activation[15,16,17]. PIK3CA resides in the 3q26 cytoband, a region frequently amplified (10-30%) in diverse cancers including squamous carcinomas of the lung, cervix, head and neck, and esophagus, and in serous ovarian and uterine cancers[11,12].

Potential relevance: The PI3K inhibitor, alpelisib[18], is FDA approved (2019) in combination with fulvestrant for the treatment of patients with PIK3CA-mutated, hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative, advanced or metastatic breast cancer. Additionally, a phase Ib study of alpelisib with letrozole in patients with metastatic estrogen receptor (ER)-positive breast cancer, the clinical benefit rate, defined as lack of disease progression ≥ 6 months, was 44% (7/16) in PIK3CA-mutated tumors and 20% (2/20) in PIK3CA wild-type tumors[19]. Specifically, exon 20 H1047R mutations were associated with more durable clinical responses in comparison to exon 9 E545K mutations[19]. However, alpelisib did not improve response when administered with letrozole in patients with ER+ early breast cancer with PIK3CA mutations[20]. Case studies with MTOR inhibitors sirolimus and temsirolimus report isolated cases of clinical response in PIK3CA mutated refractory cancers[21,22].

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## Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

### PIK3CA p.(V105_R108del) c.314_325delTAGGCAACCGTG

| Relevant Therapy | FDA | NCCN | EMA | ESMO | Clinical Trials* |
|---|---|---|---|---|---|
| inavolisib, palbociclib, hormone therapy | ✖ | ✖ | ✖ | ✖ | ● (III) |
| inavolisib | ✖ | ✖ | ✖ | ✖ | ● (II) |
| hormone therapy, alpelisib | ✖ | ✖ | ✖ | ✖ | ● (I) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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## Clinical Trials in Taiwan region:

## Clinical Trials Summary

### PIK3CA p.(V105_R108del) c.314_325delTAGGCAACCGTG

| NCT ID | Title | Phase |
|---|---|---|
| NCT04191499 | A Phase III, Randomized, Double-Blind, Placebo-Controlled Study Evaluating the Efficacy and Safety of Inavolisib Plus Palbociclib and Fulvestrant Versus Placebo Plus Palbociclib and Fulvestrant in Patients With PIK3CA-Mutant, Hormone Receptor-Positive, Her2-Negative, Locally Advanced or Metastatic Breast Cancer | III |
| NCT04188548 | EMBER: A Phase Ia/Ib Study of LY3484356 Administered as Monotherapy and in Combination With Anticancer Therapies for Patients With ER+ Locally Advanced or Metastatic Breast Cancer and Other Select Non-Breast Cancers | I |
| NCT04589845 | Tumor-Agnostic Precision Immunooncology and Somatic Targeting Rational for You (TAPISTRY) Phase II Platform Trial | II |

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## Signatures

**Testing Personnel:** Shu-Ying Wang 2022.12.30 15:00:08 +08'00'

**Laboratory Supervisor:** 2022.12.30 15:00:23 +08'00'

**Pathologist:** Yi-Chu Yeh 2022.12.30 15:00:47 +08'00'

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